

Quiz 2 Section BCS-4F

Name M

RollNo:

Short Questions (10 Marks)

1. Differentiate between process and thread

Process: A process is an independent process having its own memory.

Thread: A thread is a part of a program which shares memory.
Shares memory.

2. Explain any three advantages of multithreading with examples.

Advantages of multithreading:

- ① As the threads share memory of the process, they do efficient communication.
- ② They run parallelly, so they perform better execution of a task.
- ③ Threads are better as they manage resources themselves.

Example: As threads run in parallel, so if one ^{get} stuck in I/O, the other threads would run smoothly.

3. Explain the difference between:

- User-level threads
- Kernel-level threads

Also give one advantage and one disadvantage of each.

User-Level threads:

Threads are created and managed by the user. The kernel doesn't have any info. about them.

Advantage: The context-switching is much faster in threads.

Disadvantage: threads are parallel running so blockage can occur.

Kernel-Level threads:

Threads are created and managed by the kernel. The kernel treats each thread as a separate entity.

Advantage: Each thread runs on a different CPU core.

Disadvantage: Slower context switching as kernel switches them.

MCQs (10 Marks)

1. A thread is considered:

- A) Heavyweight B) Kernel only entity C) Independent process D) Lightweight

2. Which of the following is TRUE about threads?

- A) Each thread has its own memory
B) Threads share process memory
C) Threads cannot communicate
D) Threads are slower than processes

3. Context switching is faster in:

- A) Processes B) Threads C) Both same D) None

4. Which library is used for pthreads in C?

- A) stdio.h B) stdlib.h C) pthread.h D) threadlib.h

5. What does pthread_create() return on success?

- A) 1 B) -1 C) 0 D) Thread ID

6. What is the purpose of pthread_join()?

- A) Create thread B) Terminate thread C) Wait for thread completion D) Pause execution

7. In user-level threads:

- A) OS manages threads B) Kernel manages threads
C) OS is not aware of threads D) Threads run in kernel mode

8. A disadvantage of user-level threads is:

- A) Slow execution B) High memory usage
C) If one thread blocks, all block D) No concurrency

9. Which system call is used to create a FIFO in C?

- A) pipe() B) mkfifo() C) fifo() D) create()

10. Which statement is correct?

- A) Processes share memory B) Threads are independent
C) Threads share resources D) Threads cannot run concurrently